

Subject Code	AF5353
Subject Title	Security Analysis and Portfolio Management
Credit Value	3
Level	5
Normal Duration	One Semester
Pre-requisite / Co-requisite/ Exclusion	Have good knowledge of using basic functions and commands in EXCEL
Objectives	There are two major emphases in this course. The first part of the course focuses on portfolio analysis and the second part of the course focuses on investment management process. This subject helps achieve the Outcomes by enabling students to apply theories and professional knowledge to conduct portfolio analysis with real investment problems and solve portfolio management issues (Outcome 2), and to critically examine the internal and external situations relate to investment problems (MSc AFA Programme Outcome 1). This subject also enables students to understand the roles of financial technologies (such as artificial intelligence (AI), big data, cloud computing and others) in the financial markets.
Intended Learning Outcomes	Upon successful completion of this course, students should be able to: a. Understand Risk and Return in the financial markets b. Give recommendation of investment plans based on investors' circumstance including policy statement, asset allocation strategy, mutual fund selection, and the portfolio construction c. Apply single-factor and multifactor models to construct real equity portfolios d. Evaluate the performance of equity funds with up-to-date performance measures e. Understand and evaluate how the new technologies, including artificial intelligence (AI), big data and data analytics, and cloud computing influence the development of the financial markets.

Subject Synopsis/ Indicative Syllabus	The Mean-Variance Analysis and Portfolio Optimization in Practice The issues in the use of the mean-variance optimization in practice and possible solutions for them Asset Pricing Models and Factor Models The single-factor model and multi-factor models; the correlation structures of security returns under asset pricing models; and the applications of asset pricing models in equity portfolio construction Investment Management Process Policy statement, asset allocation strategy, portfolio construction and implementation, and international issues Equity Portfolio Management Strategies Asset allocation strategies; active, passive and semi-active portfolio management strategies Portfolio Performance Evaluation and Risk Measure Holding-based portfolio performance measures; and an introduction of downside risk measures and the Value-at-Risk measure Alternative Investment and Structured Securities An introduction of alternative investments, hedge fund strategies and pricing structured securities Behavioral Finance (Optional) The impact of heuristic-driven biases on investment decision making including representativeness, overconfidence, anchoring-and-adjustment, and aversion to ambiguity
Teaching/Learning Methodology	Key concepts and techniques will be introduced through lectures. The course places a lot of emphasis on project work. Students will be required to deliver a project which emphasizes on real-world investment issues. By completing the project, students should have hands-on experience in using the knowledge they have learned in class to solve investment problems in practice. Students are encouraged to share their views and experiences actively with their lectures and classmates.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	
	1. Class Participation	5%	√	√	√	√	√	
	2. Project	35%	√	√	√	√	√	
	3. Assignment 1	10%	√	√	√			
	4. Midterm	15 %	√	√	√			
	5. Final Examination	35 %	√	√	√			
	Total	100 %						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.							
	To reflect the significant technology content in this subject, 10% (or more) of the overall weighting of this subject is based on individual assessment concerning technology-related knowledge.							
	<i>Note: The specific requirements on individual assessment components discussed above could be adjusted based on the pedagogical needs of subject lecturers.</i>							
Student Study Effort Expected	Class contact:							
	▪ Lectures / Seminars						39 Hrs.	
	Other student study effort:							
	▪ Reading materials / textbook questions						39 Hrs.	
	▪ On average around 16 hours will be spent on the individual critique and around 20 hours for the group project discussion, presentation and written report						36 Hrs.	
	Total student study effort						114 Hrs.	

Reading List and References	Reference		
	<i>Essentials of Investment</i> , 9th edition, Zvi Bodie, Alex Kane and Alan Marcus (McGraw-Hill/Irwin, 2012)		
	<i>Modern Portfolio Theory and Investment Analysis</i> , 8th edition, Edwin J. Elton, Martin J. Gruber, Stephen J. Brown, and William N. Goetzmann (John Wiley & Sons, 2011)		
	<i>Analysis of Investments and Management of Portfolios</i> , 10th edition, Frank K. Reilly and Keith C. Brown (South-Western, 2012)		
	Other Reference		
	Title	Authors	Why?
	(Other good textbooks)		
	Investment	Zvi Bodie, Alex Kane and Alan Marcus	An advanced version of <i>Essentials of Investment</i>
	Investment Science	David Luenberger	First course in Quantitative Finance (Intermediate Investment)
	Statistical Models and Methods for Financial Market	TL Lai and H. Xing	Best statistical modeling book
Algorithmic Trading and DMA	Barry Johnson	First course in algorithmic trading and orders splitting	
(Investment in Practice)			
Active Portfolio Management	Richard Grinold and Ronald Kahn	A Quantitative Approach for Providing Superior Returns and Controlling Risk	

	The Intelligent Investor	Benjamin Graham	Best book of practice in Value Investing
	A random walk down wall street	Burton G. Malkiel	Individual investors are better off buying and holding onto index funds
	A non-random walk down wall street	Andrew Lo and A. C. MacKinlay	views again the previous book
	Fooled by randomness	Nassim N. Taleb	Lucky or Skill?
	Black Swan	Nassim N. Taleb	All the swans are white?
	Irrational Exuberance	Robert J. Shiller	Internet Bubble 1998-2001, most famous word of Nobel Laureates
	(Investment Banking)		
	Monkey Business	John Rolfe and Peter Troob	Entry level iBanker's life
	Barbarians at the Gate	Bryan Burrough and Johb Helyar	M&A classic book (usually higher level)

Reading list and references	(1). Bratko, I., 2001, PROLOG, Programming for Artificial Intelligence, 3rd edition, Addison-Wesley. (2). Luger, G.F., 2009, Artificial Intelligence: Structures and Strategies for Complex Problem Solving, 6th edition, Addison-Wesley. (3). Russell, S. and Norvig, P., 2003, Artificial Intelligence - A Modern Approach, 2nd edition, Prentice Hall. <i>Papers and articles selected from:</i> Artificial Intelligence AI Expert AI Magazine Applied Intelligence IEEE Computer IEEE Intelligent Systems and their Applications IEEE Trans. Neural Networks
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